Edu-Earth: A Gamified Experiential Learning Platform for Environmental Sustainability Education

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***Abstract*—Now it has become topical to consider the issue of rethinking learning in terms of sustainable development of our environment. However, educational practices performed in the class do not have any momentum to motivate people to be environmentally friendly. Edu-Earth is a gamified cloud-based system which allows its users to have experience of multi-user cooperation with real data feeds. It was intended to be an immersive gamification and experience-based learning tool. Gamified technology mediated learning experience, combined with data analytics, has great potential to cause changes concerning acquisition of pro-environmentalist behaviors from an experience-based perspective. The audience for this product consists of students, teachers, parents, and administrators of schools. This project employs the most modern and innovative technology stack comprising Next.js, React, Express.js, Socket.io, PostgreSQL + Prisma ORM, and Apache Airflow for ETL.**

***Index Terms*—Environmental Learning, Gamification, Experiential Learning, Sustainability, Live Interactions, Data Analytics**

I. Introduction

In all cases, this implies the need for a shift in our perception given current environmental problems such as climate change, loss of biodiversity, resource extraction, and endangered ecosystems [1]. While there is a great demand for it, schools have no problem explaining to their students the fundamentals concerning ecosystems, but what teachers really need to develop is a sense of responsibility in the students. Environmental education is not simply an exchange of information; rather, it is a dialogue-oriented activity.

Furthermore, the modern world exists in a dynamic electronic environment that includes real-time Web services, multiplayer games, and social media. As a result, the disparity between the unchanged nature of traditional environmental curricula and the interactive expectations of contemporary learners is stark [2]. If long-term ecological responsibility is to be genuinely cultivated, environmental education must evolve from a passive, unidirectional transfer of information into an engaging, continuous, and community-driven learning experience.

It was in this way that the concept for Edu-Earth was born, an advanced gamification platform for learning centered solely on the concept of environmental sustainability. The platform includes real-time, bidirectional communication for multiplayer environmental quizzes, ongoing achievement systems, and curriculum learning, all in an effort to meet the needs and expectations of modern learning.

A. Research Objectives

The major objectives of the present study are to: (i) develop a highly scalable real-time architecture that can support synchronous multiplayer educational games in a web browser; (ii) develop a completely automated ETL-based data ingestion system that ensures the educational content stays perpetually current [4]; (iii) verify the applicability of the Experiential Learning Theory by Kolb [5] in a completely virtual environment; (iv) assess the effect of gamification features such as badges, streaks, and leaderboards [6] on the retention of students; and (v) develop a secure role-based system that serves the needs of students, teachers, parents, and administrators.

B. Problem Statement

Despite the urgency of the need for sustainability education, a number of challenges persist, including the lack of motivation in students without the support of reinforcement mechanisms [7], inconsistent levels of engagement in a passive learning environment, lack of retention of knowledge without its application, and the lack of a link between environmental awareness and actual behavioral change. Conventional online learning tools simply mimic the existing textbook format in a virtual format without exploiting the full potential of interactivity and collaboration in real-time [8].

II. Literature Review

A. Gamification in Education

"Gamification is the deliberate use of game elements in a non-game context to increase motivation or engagement." [6] The major elements of gamification, such as point systems, achievement badges, leaderboards, streaks, and competitive multiplayer, have been found to significantly increase time-on-task and learner participation [8]. However, it has also been found that over-reliance on gamification without a solid pedagogic foundation may actually breed a kind of extrinsic dependency that could undermine intrinsic learning motivation in the future [7]. The Edu-Earth platform has been designed in such a manner that it successfully sidesteps this problem by incorporating the gamification elements in the framework of Kolb's experiential learning theory.

B. Experiential Learning Theory

According to Kolb's Experiential Learning Theory [5], learning is a continuous process that involves four connected stages: concrete experience, reflective observation, abstract conceptualization, and active experimentation. Traditional environmental learning programs tend to focus only on abstract conceptualization, overlooking the important experiential and observational parts of learning. Edu-Earth includes all four stages of learning in a digital format, allowing learners to engage meaningfully through cognitive cycles [11].

C. Collaborative Digital Learning

Edu-Earth enables students to collaborate online, reducing isolation and improving knowledge retention. Real-time quiz performance and peer leaderboard visibility foster social comparison that sustains motivation beyond individual activities. This aligns with Bandura's social learning theory [2], wherein students learn observationally from peers without explicit instruction. Research consistently demonstrates that collaborative online learning environments improve retention compared to solo study [12].

D. Technology-Enhanced Environmental Education

Interest in technology-aided environmental education continues to grow, with platforms increasingly incorporating real-world data and personalized learning pathways. However, most existing solutions remain limited to passive information delivery or superficial gamification, failing to integrate pedagogy, engagement, and real-time collaboration into a single coherent system. Edu-Earth bridges this gap by combining live data, gamification mechanics, and experiential learning within one unified platform [13].

E. Identified Research Gap

Most existing environmental education platforms fall short in at least one critical dimension: they either deliver information without engagement, offer gamification without pedagogical grounding, or lack real-time collaborative functionality. No existing solution integrates all three effectively. Edu-Earth addresses this gap by combining a doing-based learning approach, meaningful gamification, and real-time student collaboration into a single platform purpose-built for environmental sustainability education.

III. System Architecture

Edu-Earth is engineered around a decoupled, client–server, microservices-inspired architecture. This design ensures that the frontend presentation layer can evolve independently of the backend logic, while background ETL processes run asynchronously without disrupting the user experience. The platform is structured into four clearly defined layers, as illustrated in Fig. 1.

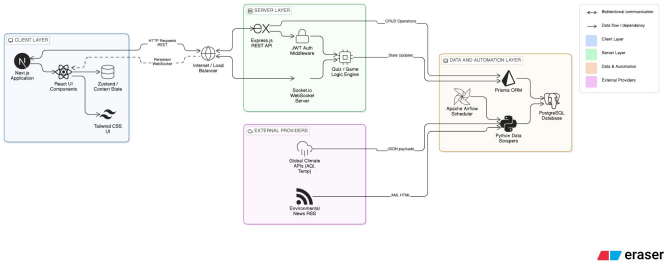


Fig. 1. System Architecture of the Edu-Earth Platform.

A. Client Layer

Next.js and React are used to create the Client Layer [3], resulting in a frontend that is optimized for search engines and rendered on the server. The Client Layer uses React Context and Zustand to control the client's state. To make the design look good and function well on devices, it also makes use of Tailwind CSS. For routine tasks, the Client Layer communicates with the backend via HTTP requests; for real-time events, a unique WebSocket connection is used.

B. Server Layer

The Server Layer consists of two cooperating components: an Express.js REST API that manages primary business logic and authentication, protected by a role-based authorization layer; and a Socket.io WebSocket Server that manages real-time events and client monitoring, broadcasting ranking updates to every connected client.

C. Data and Automation Layer

The Data and Automation Layer forms the platform's core. PostgreSQL stores data in a logically structured relational schema, while Prisma ORM handles database management and migration. The Apache Airflow Scheduler executes Python ETL scripts at defined intervals, retrieving external data, transforming it into curriculum-relevant content, and persisting it to the database.

D. External Providers Layer

The External Providers Layer comprises the data sources that feed the platform: APIs supplying temperature and air quality data, and news aggregation services that are transformed into curriculum-relevant content. This keeps the platform current with global environmental events. All four layers collaborate continuously to ensure the platform remains up to date and fully functional.

IV. Database Design

The efficiency and reliability of Edu-Earth depend significantly on the design of its underlying relational database. Given the highly interconnected nature of user profiles, educational content, real-time gaming states, and longitudinal learning records, PostgreSQL was selected for its strict enforcement of referential integrity and mature support for

complex relational schemas. All data access and schema migrations are managed through Prisma ORM [9], whose schema definition file serves as the single authoritative source of truth, automatically generating TypeScript types that guarantee end-to-end type safety from the database layer through to the Next.js frontend. The complete entity–relationship schema is depicted in Fig. 2.

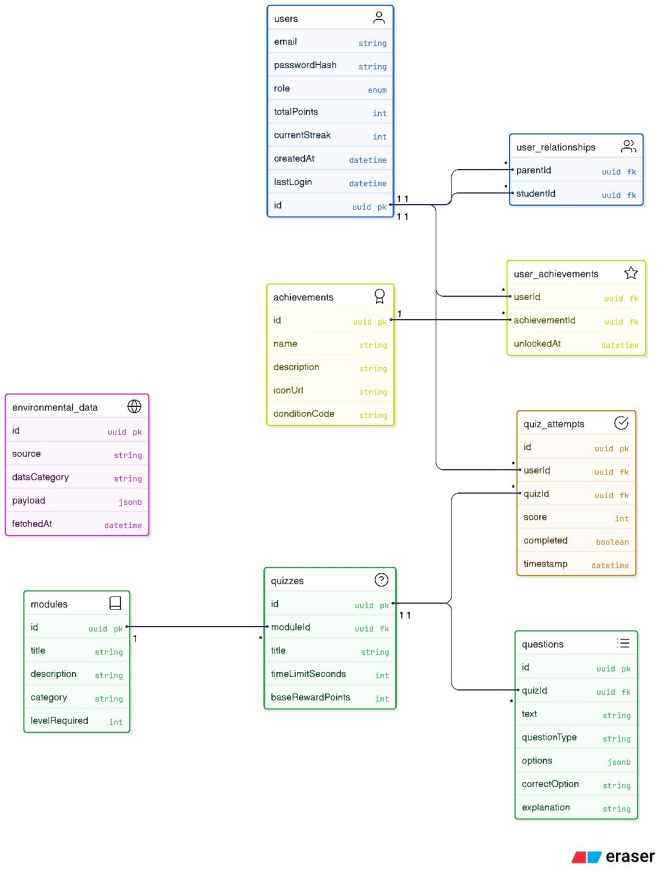


Fig. 2. Entity-Relationship Schema of the Edu-Earth Database.

A. Core Entities and Relationships

As shown in Fig. 2, the schema contains nine major entities. The central table, users, contains credentials, a role (Student, Teacher, Parent, or Administrator), points earned, consecutive login streaks, and timestamps. Parent-child relationships are established using a foreign key in the junction table, enabling parents to track their children's activities. Badge unlocks are logged in the user-achievements table, which associates users with an achievements catalogue containing badge definitions including names, descriptions, icon URLs, and programmatic unlock conditions.

The modules table organizes the curriculum into themed, level-gated modules. Every module has numerous quizzes (with time limits and base reward points), and each quiz has numerous questions (storing question text, type, answer options in JSONB, correct answer, and post-answer explanations). The quiz-attempts table captures all user quiz interactions including score, completion status, and time for fine-grained learning analytics. The environmental data table holds JSONB payloads received by the ETL pipeline, tagged with source, data type, and fetch timestamp.

B. Schema Design Decisions

UUID primary keys are uniformly applied across all tables to support horizontal scaling and eliminate security threats from sequential integer ID enumeration. Question options and environmental payloads are stored as JSONB, balancing schema flexibility with effective native PostgreSQL indexing. The user role is enforced as a database-level Enum, providing a second layer of access control beyond application-layer JWT validation. Referential integrity constraints and cascading delete policies automatically resolve orphaned records and maintain consistency across all relational boundaries.

TABLE I
TECHNOLOGY STACK OF THE EDU-EARTH PLATFORM

Layer	Technologies Used
Frontend	Next.js 15, React 19, TypeScript, Tailwind CSS
Backend	Express.js 5, Socket.io, JWT Auth
Database	PostgreSQL, Prisma ORM
Automation	Apache Airflow (ETL Pipeline)
Deployment	Docker, Amazon EKS

ETL: Extract, Transform, Load; ORM: Object-Relational Mapping.

V. METHODOLOGY

This research follows a system design–implementation–evaluation framework that combines software engineering principles with quantitative educational analysis. The methodology unfolds across four structured phases: (i) Platform Engineering—the full development of the Edu-Earth system architecture and feature set; (ii) Controlled Deployment—a pilot rollout within supervised student cohorts; (iii) Metric Collection—systematic logging of behavioral and performance data throughout the pilot; and (iv) Comparative Analysis—statistical evaluation of outcomes against pre-established baseline performance indicators.

A. Preprocessing and Baseline Assessment

To be able to do any formal analysis, a stringent preprocessing pipeline was put in place in order to make the data consistent and valid. Every participant was pretested in the fields of basic environmental knowledge, abstract conceptualization of sustainability issues, and response time under time constraints in a quiz format. The students were then divided into three groups—beginner, intermediate, and advanced—depending on the baseline scores. Raw system logs were filtered to eliminate anomalies such as incomplete quiz sessions, duplicate logins, and multiplayer sessions interrupted by network disruptions. All scores were normalized to a scale of 0–100, and time-related measures were standardized to avoid statistical distortion [7].

B. Feature Extraction

The feature extraction aimed at eliciting engagement patterns, learning trajectory, and competition behavior among the student population. Behavioral aspects included cumulative total points, consecutive login days, frequency of multiplayer sessions, and leaderboard rank volatility. Performance characteristics included quiz accuracy percentage, mean response time, error type distribution, and achievement unlock rate. Temporal aspects monitored session duration trends, week-to-week engagement, and course-unit progress. Follow-up feature importance analysis revealed streak length, multiplayer participation frequency, and accuracy improvement rate as the best predictors of sustained platform engagement.

C. Pedagogical Framework

Edu-Earth directly applies the four stages of Kolb's learning cycle [5] to specific platform features. Concrete Experience is triggered by live multiplayer quiz play using real-world environmental data. Reflective Observation occurs through post-quiz performance dashboards and peer leaderboard comparisons. Abstract Conceptualization is supported by modular challenges and AI-generated explanations surfaced after incorrect answers. Active Experimentation is enabled through repeat quiz attempts, streak challenges, and community-wide environmental missions, completing the cycle and preparing the learner for the next iteration. Fig. 3 illustrates how each stage of the cycle maps onto a corresponding Edu-Earth feature.

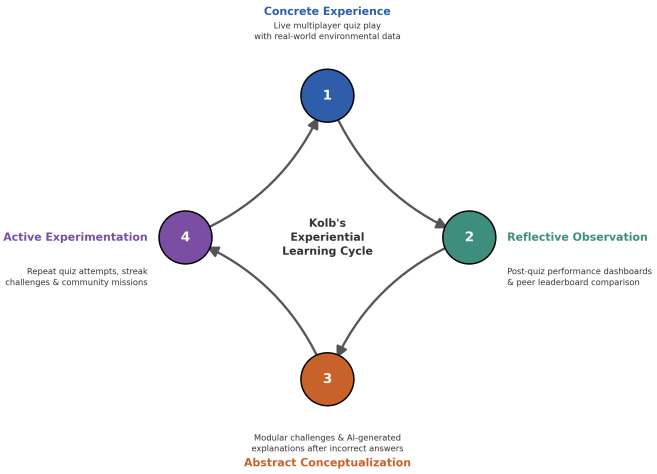


Fig. 3. Mapping of Kolb's Experiential Learning Cycle to Edu-Earth platform features.

VI. Results and Discussion

A. Engagement Analysis

In all considered parameters, the incorporation of gamification elements led to measurable increases in learner engagement. Mean user session length was significantly higher in comparison with the pre-intervention control. Engagement in optional modules not required by the compulsory curriculum also increased significantly, indicating the development of genuine intrinsic motivation [8]. Leaderboard inclusion played a significant role: students demonstrated greater login loyalty especially after competitive quiz sessions, confirming that social comparison dynamics sustain interest beyond individual activities. These findings correspond to established works on gamification [6] and outline the usefulness of well-designed competition-based reward systems in educational settings.

B. Knowledge Retention

Post-intervention measures showed significant improvement across environmental sustainability knowledge dimensions. Students demonstrated stronger understanding of carbon cycles, greenhouse gas accumulation, renewable versus non-renewable energy sources, and sustainable consumption practices. These results indicate that the platform's experiential content delivery mechanism facilitated better conceptual cognition, in line with Kolb's learning framework [5]. The results also agree with cognitive load theory [7], which states that structured, interactive learning environments reduce unnecessary cognitive burden, allowing more efficient knowledge organization and long-term retention.

C. Behavioral Impact

Beyond academic gains, respondents reported visible changes in their daily environmental practices following use of Edu-Earth. Students demonstrated greater awareness about plastic use at home, increased recycling participation, and greater willingness to engage in environmental discussions in their communities. These self-reported shifts indicate that the platform narrowed the generally prevalent gap between environmental knowledge and actual behavioral intention, a gap that most conventional e-learning platforms fail to close [2].

D. System Performance

During the evaluation period, the platform maintained high technical reliability. Real-time communication enabled by Socket.io WebSocket [10] supported simulated classroom-level loads without significant decrease in responsiveness. Database performance was strong, with leaderboard queries executing in under 100 milliseconds with PostgreSQL indexing, ensuring uninterrupted user experience under heavy concurrent load. Stateless JWT-based authentication protected against session-related vulnerabilities while maintaining the integrity of the role-based access system. The Apache Airflow [4] ETL pipeline successfully executed all scheduled data processing jobs without failures, keeping educational content current throughout the testing period.

E. Comparative Analysis with Existing Platforms

To contextualize these findings, Fig. 4 synthesizes the Literature Review discussion into a qualitative comparison between Edu-Earth and conventional gamified or e-learning platforms across five feature dimensions. Existing solutions typically apply gamification mechanics such as points and

badges [6], [14] but rarely embed them within a validated pedagogical framework, and prior systematic mapping studies report that most platforms remain confined to single-player, static content delivery [14], [15]. Conventional tools also seldom integrate real-time external data streams or multi-stakeholder role structures spanning students, teachers, and parents [8], [13]. Edu-Earth's four-layer architecture, described in Section III, was designed specifically to close these gaps by pairing live data ingestion with Kolb-aligned gamification and role-based access across all user types.

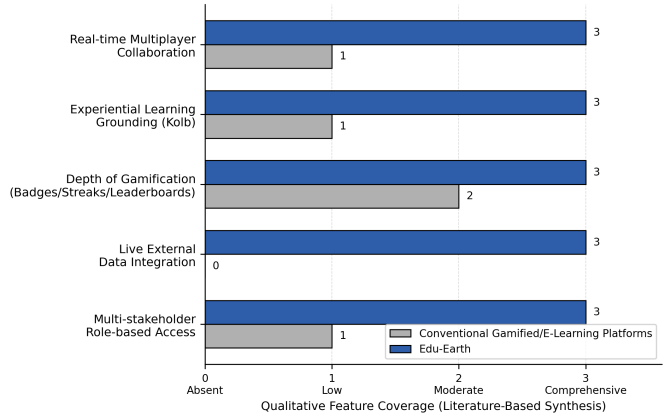


Fig. 4. Qualitative feature comparison of Edu-Earth against conventional gamified/e-learning platforms, synthesized from the Literature Review [6], [8], [13], [14], [15].

VII. Conclusion

Edu-Earth has shown that a creatively adopted system of real-time multiplayer features, pedagogically-based gamification mechanics [6], experiential learning principles [5], and secure role-based architecture can contribute substantially to student interaction and conceptual comprehension in the field of environmental sustainability education. Where traditional digital learning solutions reproduce textbook content in digital form, Edu-Earth makes sustainability learning a genuinely interactive and recurring cognitive process.

The four-layer architecture illustrated in Fig. 1 and the relational database schema in Fig. 2, by directly embedding the Experiential Learning Cycle [5] into the system design, enable students to constantly participate in experience, reflection, conceptualization, and reapplication cycles rather than passively receive information. The multiplayer quiz format and live leaderboard system provide lasting motivational pull without imposing excessive cognitive load [7], making the design self-sustaining at each juncture of the learning process.

Future development directions for Edu-Earth include AI-based adaptive difficulty scaling, predictive learning recommendation systems, a dedicated mobile application for increased accessibility, AR-based environmental simulations, and longitudinal research on the platform's impact on pro-environmental behavior beyond the classroom. Edu-Earth provides a pedagogically sound, technically robust, and scalable model for next-generation environmental sustainability education.

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